

SafeByte File System Crawler

User Guide — V2.0

Malware Detection · VirusTotal Integration · MD5/SHA256 Hashing

The SafeByte File System Crawler scans any directory on your computer, hashes every file, checks it against VirusTotal's database of 70+ antivirus engines, and flags suspicious or malicious files using local heuristics — all in a simple GUI application that requires no installation.

1. Installation

macOS

- Unzip **SafeByte-File-Crawler-macOS.zip**.
- Drag **SafeByte File Crawler.app** to your Applications folder.
- Right-click → Open on first launch to bypass Gatekeeper.

Windows

- Run **SafeByte File Crawler.exe** directly — no installation needed.
- If Windows Defender blocks it: click **More info** → **Run anyway**.

■■ The app requires an internet connection for VirusTotal lookups. Local heuristics work offline.

2. Interface Overview

The application window has five main areas:

Area	Description
Directory Bar	Shows the selected folder path. Use Browse... to change it.
Control Buttons	Start Scan, Stop, Export CSV, Export TXT.
Stats Bar	Live counters: Total Files, Safe, Suspicious, Malicious, Skipped.
Progress Bar	Animated while scan is running. Shows current file being scanned.
Results Table	Every scanned file with verdict, hash, VT score, and reasons.

3. Running a Scan

Step 1 — Select a Directory

- Click **Browse...** or type a path directly in the directory bar.
- Default is your home folder. For a full system scan, enter `/` on macOS or `C:\` on Windows.
- Note: scanning `/` or `C:\` can take hours and generate thousands of results.

Step 2 — Configure VirusTotal (optional)

- The **VirusTotal hash lookup** checkbox is enabled by default.
- When enabled, every file's SHA256 hash is checked against VirusTotal's database.
- This is slower (free tier: 4 requests/minute) but catches known malware.
- Disable it for fast local-only scans.

Step 3 — Start Scan

- Click  **Start Scan**. The progress bar animates and the status bar shows the current file.
- Results appear in the table in real-time as each file is processed.
- Click  **Stop** at any time to halt the scan. Results already found are kept.

 **Scanning system directories (`/System`, `/Windows`) may produce permission errors. These files are automatically marked as **Skipped**.**

4. Understanding Verdicts

Each file receives one of four verdicts:

Verdict	Colour	Meaning
MALICIOUS	Red	VirusTotal flagged this file as malware by one or more engines.
SUSPICIOUS	Amber	Local heuristics matched (dangerous filename, extension, or location). Not confirmed malware.
SAFE	Green	No threats detected by local rules or VirusTotal.
SKIPPED	Grey	File too large (>50MB) or permission denied. Could not be analysed.

Local Heuristics — What Gets Flagged

- **Dangerous filenames:** `mimikatz`, `meterpreter`, `keylogger`, `wannacry`, `payload`, `backdoor`, etc.
- **Suspicious extensions:** `.exe`, `.dll`, `.bat`, `.ps1`, `.vbs`, `.sh`, `.jar`, `.msi`, `.docm` (macro Office), disk images, etc.
- **Suspicious locations:** `/tmp/`, `AppData\Roaming\Startup`, `LaunchAgents`, cron directories, etc.

■■ SUSPICIOUS verdict does NOT mean the file is malware. Many legitimate files have .exe or .sh extensions. Always check the Reasons column and use your judgement.

5. Results Table Columns

Column	Description
Verdict	MALICIOUS / SUSPICIOUS / SAFE / SKIPPED
Filename	Name of the file only (without path)
Full Path	Complete absolute path to the file
Ext	File extension
Size (KB)	File size in kilobytes
MD5	MD5 hash of the file contents
VT Malicious	Number of VirusTotal engines that flagged this file
VT Total	Total number of VirusTotal engines that analysed this file
Reasons	Explanation of why the file was flagged

Click any column header to sort the table by that column.

6. Exporting Results

Export CSV

- Click ■ **Export CSV** after the scan completes.
- Saves all results including hashes, VT scores, and reasons.
- Open in Excel, Google Sheets, or any spreadsheet tool for analysis.

Export TXT

- Click ■ **Export TXT** for a plain-text threat report.
- Only MALICIOUS and SUSPICIOUS files are included — clean files are omitted.
- Useful for sharing with an IT security team or keeping as an audit record.

7. Tips & Best Practices

- Start with your **Downloads** and **Desktop** folders — common malware drop zones.
- Check **AppDataRoaming** (Windows) and **~/Library/LaunchAgents** (macOS) for persistence.
- If VT returns 1–2 detections on a common system file, it may be a false positive.
- Cross-reference MALICIOUS files on **virustotal.com** before deleting anything.

- Never delete a SUSPICIOUS file just because of extension — verify the Reasons first.
- Run scans periodically (weekly) to catch newly downloaded malware.

■■ Do NOT delete system files based solely on this tool's output. Always verify on [virustotal.com](https://www.virustotal.com) and consult IT support if unsure.